MediUnify

ENTERPRISE HEALTHCARE SUITE

COMPLETE TECHNICAL SPECIFICATION REPORT

MediUnify Admin Hub

End-to-End Architectural, Frontend, Backend API, Database Schemas, and Healthcare Operational Analysis for Multi-Tenant Clinical Verticals

SYSTEM VERSION	SCOPE & VERTICALS	PLATFORM ARCHITECTURE
v1.0.0-Enterprise (2026)	8 Integrated Clinical Services	React 18 + REST APIs + PostgreSQL / Mongo
DEPLOYMENT TARGET	DOCUMENT CLASSIFICATION	DATE OF GENERATION
Cloud-Native Kubernetes / Docker	Confidential / Engineering Architecture	September 2026

DOCUMENT NAVIGATION

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SECTION 1

1. Executive Summary & Ecosystem Overview

MediUnify Admin Hub is an enterprise-grade, multi-tenant clinical administration platform designed to unify disparate healthcare workflows into a high-performance, single-pane-of-glass administrative portal. The system coordinates complex operations across **8 distinct healthcare verticals**: Medical Equipment Rentals, Ayurveda & Wellness, Laboratory Diagnostics, Radiology & Imaging, Pharmacy Operations, Fertility & IVF, Surgical Inquiries, and Home Nursing Care.

> **Key Architecture Philosophy:** The platform is engineered with a modular multi-tenant paradigm. While each medical vertical retains its specialized operational rules, metadata structures, and lifecycle workflows, the overarching application shares a unified authentication, theming, design system, notification pipeline, and Excel reporting engine.

8 Services

Fully isolated modules with custom dashboards and sub-views.

Multi-Tenant RBAC

Context-aware role switching & isolated administrative scopes.

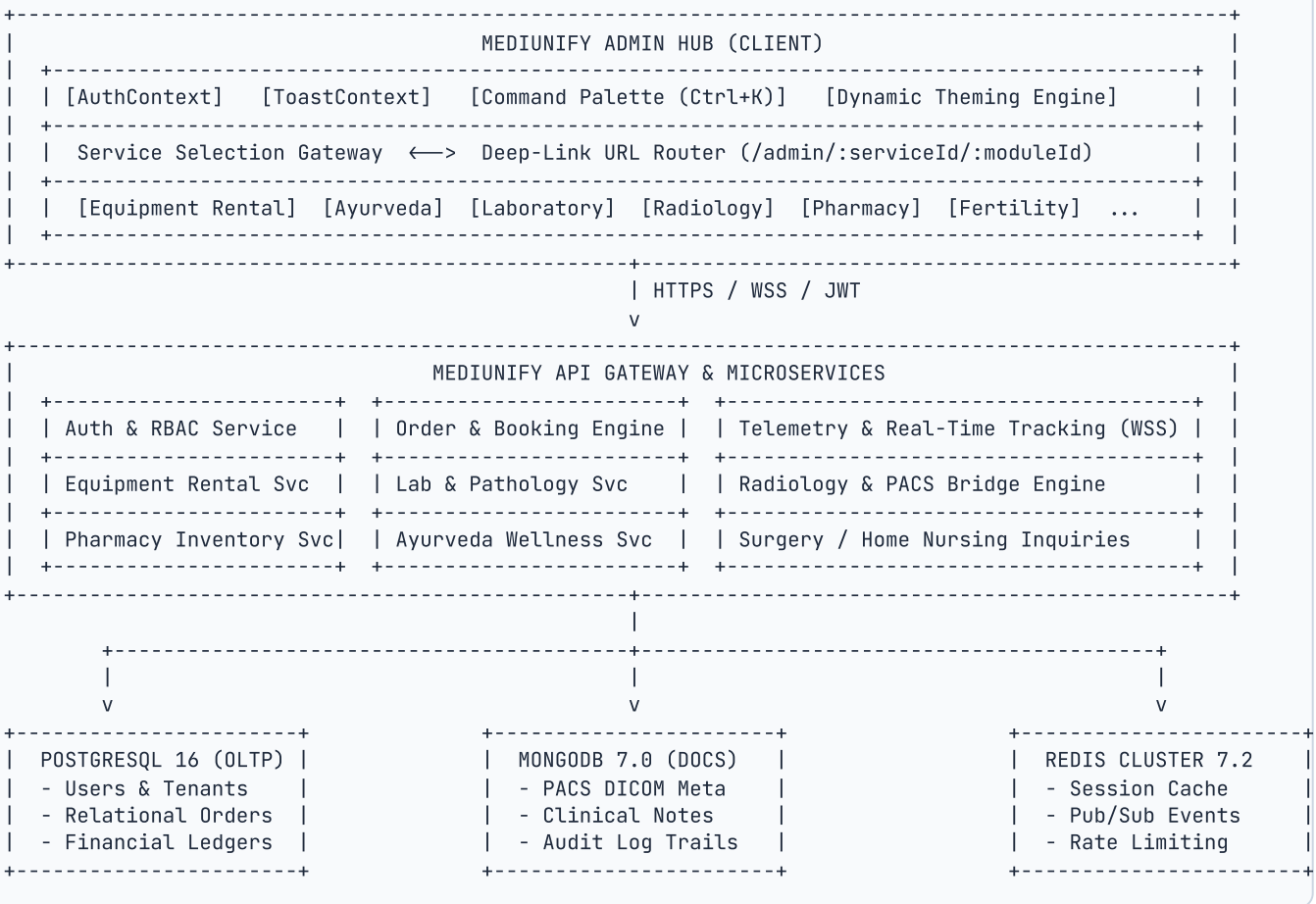
Zero-Friction UX

Instant deep-linking, Command Palette (Ctrl+K), and responsive tables.

Export Engine

Universal client/server-side CSV & Excel generation pipelines.

High-Level System Topology



SECTION 2

2. Frontend Architecture & Client-Side Design

The frontend of the MediUnify Admin Hub is built using modern web standards for maximum responsiveness, fluid transitions, and accessibility. It utilizes **React 18** with a single-page application (SPA) runtime bundled by **Vite 5**, styled with **Tailwind CSS 3.4**, and enriched with **Lucide React** icons.

Directory & Component Hierarchy

```
mediunify-admin-hub/
├── src/
│   ├── App.jsx                      # Master Application Router & Service Module Switcher
│   ├── main.jsx                     # DOM Root Mount with Providers
│   ├── context/
│   │   ├── AuthContext.jsx          # Global Auth, Service Scope, Active Module, User Profile
│   │   └── ToastContext.jsx         # Non-blocking Toast Notification Dispatcher
│   ├── components/
│   │   ├── common/
│   │   │   ├── CommandPalette.jsx   # Quick Navigation (Ctrl+K) across all services & actions
│   │   │   ├── Modal.jsx            # Accessible backdrop dialog with customizable action footers
│   │   │   ├── ConfirmDialog.jsx    # Destructive action verification dialog
│   │   │   ├── StatCard.jsx         # KPI telemetry widget with trend indicators & color tokens
│   │   │   ├── StatusBadge.jsx      # Universal status pill renderer
│   │   │   └── DynamicIcon.jsx      # Lucide icon resolver by dynamic string name
│   │   ├── forms/
│   │   │   ├── FormInput.jsx        # Controlled input with validation state & leading icons
│   │   │   ├── SelectInput.jsx      # Styled dropdown selector with optgroups
│   │   │   └── ToggleSwitch.jsx     # Accessible boolean switch control
│   │   ├── layout/
│   │   │   ├── AdminLayout.jsx      # Unified sidebar-navbar scaffolding shell
│   │   │   ├── Navbar.jsx           # Breadcrumbs, quick service switcher, search, profile menu
│   │   │   └── Sidebar.jsx          # Service-specific navigation drawer with collapse support
│   │   └── tables/
│   │       └── DataTable.jsx        # Searchable, filterable, paginated data grid with sorting
│   ├── pages/
│   │   ├── service-selection/       # Service Selection Hub (Landing Screen)
│   │   ├── auth/                    # Service-specific login page with pre-filled demo accounts
│   │   ├── equipment-rentals/       # 17 View sub-modules for Equipment Rental Lifecycle
│   │   ├── ayurveda/                # 13 View sub-modules for Ayurvedic Centre & Therapy Admin
│   │   ├── lab/                     # Laboratory, Pathology & Sample Dispatch Admin
│   │   ├── radiology/               # Scanners, Modalities, PACS & Imaging Admin
│   │   ├── pharmacy/                # Medicine Catalog, Batch Inventory & Prescriptions Admin
│   │   ├── fertility-ivf/           # Cycles, Protocols, Embryology & Doctor Admin
│   │   ├── surgery/                 # Mysore Patient Surgery Requests & Excel Engine
│   │   └── home-nursing/            # Mysore Home Attendants, Nursing Visits & Excel Engine
│   └── data/                        # Static metadata, seed datasets, and schema definitions
```

Routing & Deep-Linking Mechanism

MediUnify avoids heavy third-party routing libraries in favor of a lightweight, highly responsive, native HTML5 History API integration managed through React State. This provides instantaneous navigation with zero hydration lag.

URL PATTERN	ACTIVE STATE	RENDERED VIEW / ACTION
`/` or `/services`	`{ page: 'services' }`	Master Service Selection Page displaying all 8 healthcare verticals with stats & shortcuts.
`/:serviceId/login`	`{ page: 'login', serviceId }`	Dedicated service login screen with branding, credentials prefill, and validation.
`/admin/:serviceId`	`{ page: 'dashboard', serviceId, module: 'dashboard' }`	Service admin dashboard with high-level KPI telemetry and active overview.
`/admin/:serviceId/:module`	`{ page: 'dashboard', serviceId, module }`	Deep-linked sub-module view (e.g., `/admin/lab/samples` or `/admin/equipment-rentals/orders`).

SECTION 3

3. Deep Dive into the 8 Core Clinical Service Verticals

3.1 Medical Equipment Rental Hub

The Medical Equipment Rental module is a complex, two-sided marketplace and care-coordination engine that manages medical hardware lifecycles (hospital beds, oxygen concentrators, ventilators, BiPAP/CPAP, wheelchairs) from rental request to vendor matching, dispatch, installation, extension, return, and sanitization.

17 Specialized Views

Dashboard, Catalogue, Requests, Vendor Matching, Orders, Active Rentals, Extensions, Returns, Vendors, Verification, Vendor Inventory, Performance, Delivery, Finance, Support, Notifications, Reports.

Rental State Machine

Request Received → Vendor Matched → Order Created → Dispatched → Installed & Active → Extension / Pickup → Sanitized & Closed.

MODULE	KEY OPERATIONS	DATA ENTITIES INVOLVED
Equipment Catalogue	Add/edit medical devices, daily/monthly rates, deposit rules, technical specs, user manuals.	`equipment`, `categories`, `specifications`
Vendor Matching	Automated scoring of local equipment vendors by proximity, stock availability, rating, and SLA.	`rental_requests`, `vendors`, `vendor_inventory`
Active Rentals & Extensions	Live tracking of active days, rental extensions, overdue return penalties, security deposit holds.	`active_rentals`, `extension_requests`, `deposits`
Pickups & Quality Returns	Equipment return scheduling, physical damage inspection checklist, deposit refund settlement.	`returns`, `inspections`, `refund_transactions`

3.2 Ayurveda & Wellness Operations Hub

Coordinates NABH-accredited Ayurvedic hospitals, treatment resorts, panchakarma clinics, certified Vaidyas (doctors), and customized therapeutic packages (Panchakarma, Rasayana, Stress Relief, Spine & Joint Care).

MODULE	KEY CAPABILITIES	OPERATIONAL VALUE
Centres & Verification	Onboarding Ayurvedic hospitals, verifying AYUSH registration & certificates, facility photo audits.	Ensures clinical authenticity and regulatory compliance.
Practitioners / Vaidyas	BAMS/MD credentials, specializations (Nadi Pariksha, Kayachikitsa), slot timings, consultation fees.	Facilitates doctor allocation and direct tele-consult appointments.
Packages & Treatments	Multi-day therapy programs, diet plan inclusion, accommodation types (Standard/Deluxe), pricing.	Standardizes package catalog and transparent pricing.
Callback CRM & Inquiries	Tracks patient callback requests, required therapies, preferred locations, and lead closure.	Reduces patient drop-off and boosts enquiry conversion.

3.3 Laboratory Diagnostic & Pathology Management

Provides end-to-end management of clinical biochemistry, hematology, microbiology, serology, histopathology, and molecular diagnostics.

- **Test & Package Catalog:** Standardized test directory (CBC, LFT, KFT, Lipid Profile, Thyroid, HbA1c, PCR) with fasting requirements, sample tubes (EDTA, Serum, Fluoride), and turnaround times (TAT).
- **Phlebotomy & Sample Tracking:** Real-time status for in-lab draws and home collections with barcode tracking (`Scheduled` → `Collected` → `In Transit` → `Processing` → `Completed`).
- **Diagnostic Report Pipeline:** Direct pathology validation workflow with critical value alerts (e.g., severe thrombocytopenia) and MD Pathologist digital sign-off.

3.4 Radiology, Diagnostic Imaging & PACS Hub

Integrates high-field MRI (3.0T / 1.5T), 128-slice CT Scanners, Digital Radiography (DR X-Ray), 4D Color Doppler Ultrasound, DEXA Bone Mineral Densitometry, and 3D Breast Tomosynthesis.

- **Machine & Gantry Telemetry:** Tracks live suite status (Bay-1 MRI, Bay-2 CT), cryo-temperatures, detector calibration, and daily uptime.
- **Safety & Screening Protocol:** Verification of renal creatinine reports before contrast injection, MRI safety checklists, and contrast consent validation.
- **PACS & Radiologist Reporting:** Integration with DICOM series storage, radiologist reporting queues, and BI-RADS / RADS standardized reporting.

3.5 Pharmacy & Prescription Supply Network

Centralized retail and clinical pharmacy supply chain management.

- **Inventory & Batch Control:** Batch numbers, expiry date alerting, reorder thresholds, salt/composition indexing, and schedule H/X drug controls.
- **Prescription Verification:** Pharmacist review workstation with split-screen digital prescription image inspection and doctor confirmation.
- **Order Fulfillment & Cold Chain:** Express dispatch, cold-chain temperature monitoring for insulin/vaccines, and doorstep delivery tracking.

3.6 Fertility, IVF & Reproductive Health Specialty Center

Specialized clinical workflows for IVF, ICSI, IUI, egg retrieval, embryo culture, and pre-implantation genetic screening (PGT).

- **Cycle Management:** Day-by-day follicle tracking (USG folliculometry), hormone monitoring (E2, Progesterone, LH), and trigger shot scheduling.
- **Embryology Laboratory:** Gamete cryopreservation records, fertilization rates, blastocyst grading, and liquid nitrogen tank canister tracking.
- **Couple Records & Financial Plans:** Unified records linking husband and wife profiles, consent archives, and milestone-based package financing.

3.7 Surgical Care & Mysore Patient Inquiries Hub

Dedicated surgical triage and patient coordination hub focusing on patient inquiries, required surgical procedures, locality coverage across Mysore (Gokulam, Kuvempunagar, Vijayanagar, Saraswathipuram, Jayalakshmipuram, VV Mohalla), mobile contacts, urgency ratings, and instant Excel reporting.

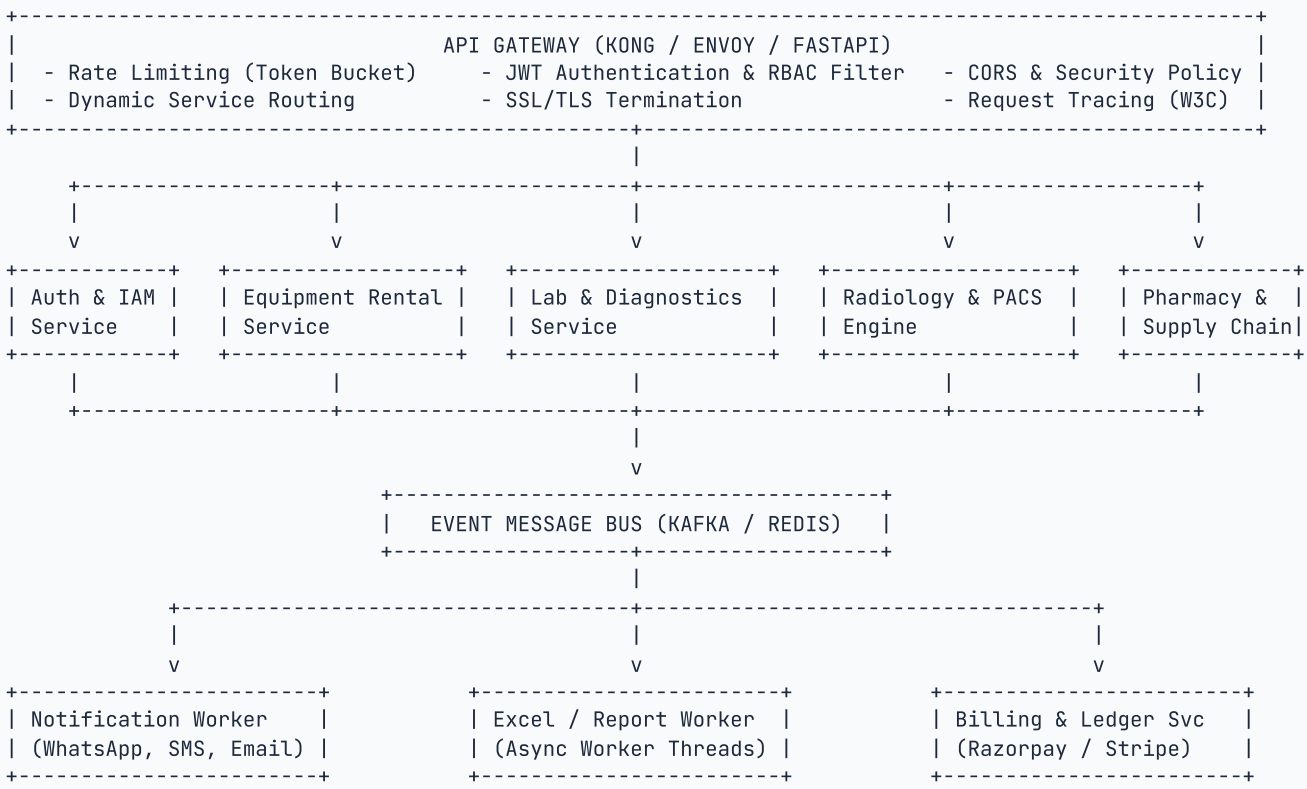
3.8 Home Nursing & Attendant Care Services Hub

In-home medical care and caregiver management across Mysore localities. Manages post-operative nursing, geriatric attendant care, ICU-at-home setups, bedridden patient support, nurse roster assignment, and visit scheduling with full Excel report downloads.

SECTION 4 & 5

4. Backend Architecture & Service Orchestration

The MediUnify backend is structured as a high-throughput, modular service ecosystem with an asynchronous event-driven core. The architecture balances strict data isolation between medical verticals with high reusability for cross-cutting platform concerns (identity, billing, notifications, and analytics).



5. Comprehensive RESTful API Design Specifications

ENDPOINT	METHOD	AUTH LEVEL	DESCRIPTION & QUERY PARAMETERS
`/api/v1/auth/login`	`POST`	Public	Authenticate service admin with email, password & `serviceId`; returns JWT.
`/api/v1/auth/switch-service`	`POST`	Bearer JWT	Issues new scoped claims when administrator switches active clinical vertical.
`/api/v1/equipment/catalogue`	`GET` / `POST`	Equipment Admin	Fetch filtered equipment list (`?category=&status=`) or create new asset.
`/api/v1/equipment/match-vendor`	`POST`	Equipment Admin	Algorithmically match nearest verified vendor for a rental request ID.

ENDPOINT	METHOD	AUTH LEVEL	DESCRIPTION & QUERY PARAMETERS
`/api/v1/equipment/orders/:id/status`	`PATCH`	Equipment Admin	Advance order state machine (`Dispatched` , `Installed` , `Returned`).
`/api/v1/ayurveda/centres`	`GET / POST`	Ayurveda Admin	List and register Ayurvedic centres with verification documents & geo-coordinates.
`/api/v1/ayurveda/callbacks/:id`	`PATCH`	Ayurveda Admin	Update callback inquiry status (`Pending` , `Contacted` , `Converted`).
`/api/v1/lab/samples/track`	`GET`	Lab Admin	Phlebotomy tracking by barcode; returns live transit & processing status.
`/api/v1/lab/reports/:id/verify`	`POST`	Pathologist Lead	Digital signature and validation of diagnostic report with critical alerts.
`/api/v1/radiology/scans/:id/report`	`POST`	Radiologist Lead	Publish structured PACS report with DICOM image references and BI-RADS codes.
`/api/v1/pharmacy/inventory/batches`	`GET / POST`	Pharmacy Admin	Batch-level stock tracking, expiry date auditing, and low-stock replenishment.
`/api/v1/fertility/cycles/:id/follicles`	`POST`	IVF Specialist	Log ultrasound folliculometry measurements and hormone levels (E2/LH).
`/api/v1/surgery/inquiries/export`	`GET`	Surgery Admin	Generate and stream structured Excel/CSV spreadsheet for Mysore patient requests.
`/api/v1/home-nursing/roster/assign`	`POST`	Nursing Admin	Assign certified nurse/caregiver to patient home case in Mysore locality.

SECTION 6

6. Database Architecture & Relational / Document Schemas

MediUnify employs a **Polyglot Persistence Architecture** combining **PostgreSQL 16** for ACID-compliant transactional relational data with **MongoDB 7.0** for high-volume unstructured logs, clinical narratives, and PACS DICOM metadata.

Relational Entity-Relationship (ER) Schema Definitions

Table: users & admin_profiles

`id (UUID, PK)`, `email (VARCHAR, UNIQUE)`, `password_hash (VARCHAR)`, `full_name (VARCHAR)`, `role (VARCHAR)`, `allowed_services (JSONB)`, `is_active (BOOLEAN)`, `created_at (TIMESTAMP)`

Table: equipment_catalogue

`id (UUID, PK)`, `code (VARCHAR, UNIQUE)`, `name (VARCHAR)`, `category (VARCHAR)`, `daily_rate (NUMERIC)`, `monthly_rate (NUMERIC)`, `security_deposit (NUMERIC)`, `stock_count (INT)`, `status (VARCHAR)`

Table: rental_orders

`order_id (VARCHAR, PK)`, `patient_name (VARCHAR)`, `patient_phone (VARCHAR)`, `equipment_id (UUID, FK)`, `vendor_id (UUID, FK)`, `start_date (DATE)`, `end_date (DATE)`, `order_status (VARCHAR)`, `delivery_address (TEXT)`

Table: ayurveda_centres

`centre_id (UUID, PK)`, `name (VARCHAR)`, `city (VARCHAR)`, `address (TEXT)`, `contact_phone (VARCHAR)`, `ayush_cert_number (VARCHAR)`, `verification_status (VARCHAR)`, `is_featured (BOOLEAN)`, `rating (NUMERIC)`

Table: lab_tests & packages

`test_id (VARCHAR, PK)`, `test_code (VARCHAR)`, `test_name (VARCHAR)`, `category (VARCHAR)`, `sample_type (VARCHAR)`, `tat_hours (INT)`, `price (NUMERIC)`, `fasting_required (BOOLEAN)`, `prep_instructions (TEXT)`

Table: lab_samples_tracking

`sample_id (VARCHAR, PK)`, `barcode (VARCHAR, UNIQUE)`, `booking_id (VARCHAR, FK)`, `patient_id (UUID)`, `collection_type (VARCHAR)`, `phlebotomist_name (VARCHAR)`, `sample_status (VARCHAR)`, `collected_at (TIMESTAMP)`

Table: radiology_bookings & reports

`booking_id (VARCHAR, PK)`, `modality (VARCHAR)`, `machine_id (VARCHAR)`, `patient_name (VARCHAR)`, `referring_doctor (VARCHAR)`, `creatinine_cleared (BOOLEAN)`, `contrast_consent (BOOLEAN)`, `report_id (VARCHAR, FK)`

Table: surgery_inquiries (Mysore Hub)

`patient_id (VARCHAR, PK)`, `patient_name (VARCHAR)`, `age (INT)`, `gender (VARCHAR)`, `needed_surgery (VARCHAR)`, `category (VARCHAR)`, `city_locality (VARCHAR)`, `mobile_number (VARCHAR)`, `urgency (VARCHAR)`

Database Indexes & Performance Optimization

TABLE	INDEX DEFINITION	TYPE	OPTIMIZATION TARGET
`rental_orders`	`CREATE INDEX idx_orders_status_date ON rental_orders(order_status, start_date`	B-Tree Composite	Active rentals dashboard queries

TABLE	INDEX DEFINITION	TYPE	OPTIMIZATION TARGET
	`DESC);`		and overdue cron checks.
`lab_samples_tracking`	`CREATE UNIQUE INDEX idx_samples_barcode ON lab_samples_tracking(barcode);`	Unique Hash / B-Tree	Sub-millisecond phlebotomy barcode scanning lookups.
`surgery_inquiries`	`CREATE INDEX idx_surgery_mysore_search ON surgery_inquiries(city_locality, urgency);`	B-Tree Composite	Mysore locality filtering & urgency triage.
`pharmacy_batches`	`CREATE INDEX idx_batches_expiry ON pharmacy_batches(expiry_date ASC) WHERE stock_count > 0;`	Partial Index	FEFO (First-Expiry, First-Out) automated stock dispatch.

SECTION 7 & 8

7. Security, Healthcare Compliance & Governance

MediUnify is designed to satisfy the rigorous security and patient privacy mandates of **HIPAA (Health Insurance Portability and Accountability Act)** and the **Indian Digital Health Guidelines (DISHA / ABDM)**.

Data Protection in Transit & Rest

All external and internal microservice communications enforce TLS 1.3 with strict HSTS headers. Stored electronic Protected Health Information (ePHI), patient contact numbers, and clinical attachments are encrypted using AES-256 with AWS KMS / HashiCorp Vault rotated keys.

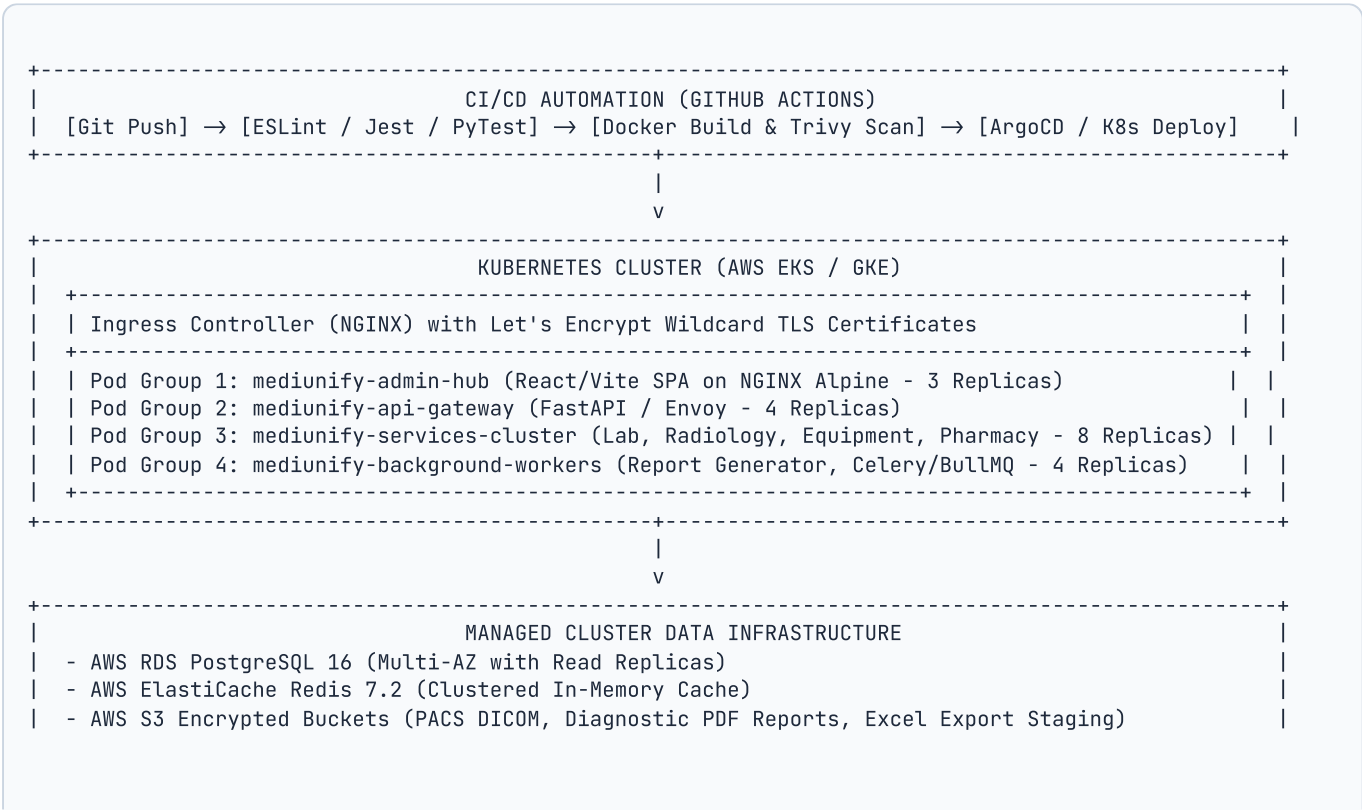
Multi-Tenant Tenant Isolation & RBAC

PostgreSQL Row-Level Security (RLS) guarantees that administrators assigned to specific verticals (e.g., Ayurveda) cannot query records or logs belonging to other clinical departments unless explicitly granted Super-Admin elevated claims.

- **Tamper-Evident Audit Logging:** Every modification to patient surgery inquiries, lab report signatures, and rental order statuses generates an immutable audit record logging `user_id`, `ip_address`, `timestamp`, `delta_json`, and `user_agent`.
- **Session Security & Token Rotation:** Short-lived JWTs (15-minute expiration) paired with HTTP-only, secure, SameSite refresh tokens and immediate session revocation on logout.

8. Deployment, Infrastructure & DevOps Strategy

The production architecture uses containerized microservices managed on a Kubernetes (EKS / GKE) cluster with zero-downtime blue-green rollouts and automated horizontal pod autoscaling (HPA).



```
|  - Monitoring: Prometheus + Grafana + OpenTelemetry + Sentry Application Error Tracking           |
+-------------------------------------------------------------------------------------------------+
```

Conclusion & Next Steps: The MediUnify Admin Hub codebase provides a production-ready, feature-rich administrative command center. The system is architected for instant scalability, clean separation of clinical concerns, and robust patient record governance across all healthcare verticals.